RQF LEVEL 4



NITIW401

NETWORKING AND INTERNET TECHNOLOGIES

IoT Web Application Development Using PHP

TRAINER'S MANUAL

October, 2024



RTB | RWANDA TVET BOARD

IoT WEB APPLICATION DEVELOPMENT USING PHP

KOICA Korea International Cooperation Agency

TQUM TVET Quality Management Project

2024

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ACRONYMS

2FA: Two-Factor Authentication

API: Application Programming Interface

API: Application Programming Interface

CBT/A: Competency-Based Training and Assessment

CD: Continuous Delivery

CSS: Cascading Style Sheets

HTML: Hypertext Markup Language

HTTP: Hypertext Transfer Protocol

IDE: Integrated Development Environment

IoT: Internet of Things

JSON: JavaScript Object Notation

KOICA: Korea International Cooperation Agency

MQTT: Message Querrying Telemetry Transport

PHP: Hypertext Pre-processor

REST: Representational State Transfer

RTB: Rwanda TVET Board

SOAP: Simple Object Access Protocol

SQL: Structured Query Language

TQUM Project: TVET Quality Management Project

UI: User Interface

WI-FI: Wireless Fidelity

XML: Extensible Markup Language

INTRODUCTION

This trainer's manual includes all the methodologies required to effectively deliver the module titled **"IoT Web Application Development Using PHP"** Students enrolled in this module will engage in practical activities designed to develop and enhance their competencies.

The development of this training manual followed the Competency-Based Training and Assessment (CBT/A) approach, offering ample practical opportunities that mirror real-life situations.

The trainer's manual is organized into Learning Outcomes, which is broken down into indicative content that includes both theoretical and practical activities. It provides detailed information on the key competencies required for each learning outcome, along with the objectives to be achieved.

As a trainer, you will begin by asking questions related to the activities to encourage critical thinking and guide trainees toward real-world applications in the labour market. The manual also outlines essential information such as learning hours, didactic materials, and suggested methodologies.

This manual outlines the procedures and methodologies for guiding trainees through various activities as detailed in their respective trainee manuals. The activities included in this training manual are designed to offer students opportunities for both individual and group work. Upon completing all activities, you will assist trainees in conducting a formative assessment known as the end learning outcome assessment. Ensure that students review the key reading and the points to remember section.

MODULE CODE AND TITLE: NITIW401 IoT WEB APPLICATION DEVELOPMENT

Learning Outcome 1: Develop Application Programming Interface (API) Using PHP
Learning Outcome 2: Develop User Interface
Learning Outcome 3: Integrate API Endpoints with User Interface


API
</>

Indicative contents
1.1 Analysis of IoT Web application requirements
1.2 Prepare PHP environment
1.3 Identification of PHP concepts
1.4 Develop Application Programming Interface CRUD Endpoints
1.5 Secure API endpoints
1.6 Test API endpoints
1.7 Documentation of the developed API

Key Competencies for Learning Outcome 1: Develop API using PHP

Knowledge	Skills	Attitudes
• Description of IoT web application requirements • Description of PHP Key terms • Description of PHP concept • Description of API concepts • Description of CRUD • Description of API end points	• Collecting data used in IoT Web application • Selecting IoT Web Application devices • Installing and configuring PHP environment tools used in IoT web application • Developing Application Programming Interface CRUD Endpoints. • Developing User Interface • Securing API endpoints • Testing API Endpoints • Documenting the developed API Endpoints and Versioning	• Being Exploratory • Having Critical thinking • Being Innovative when developing user interface • Being committed Developing Application Programming Interface CRUD Endpoints • Being confidence • Having Team work spirit • Having Adaptability

Duration: 40 hrs

Learning outcome 1 objectives:

By the end of the learning outcome, the trainees will be able to:

1. Describe properly IoT web application requirements based on organization requirements
2. Describe properly PHP Key terms, PHP key concept based on IoT system requirement
3. Describe properly API concept, API Endpoint based on IoT Web Application Development
4. Describe properly CRUD based on IoT Web Application Development requirement
5. Analyse properly data based on IoT Web application requirements.
6. Select correctly IoT Web Application devices based on system requirement.
7. Secure properly API endpoints based on PHP data security standards.
8. Test effectively API endpoints based on IoT system requirements.
9. Document properly the developed API according to its functionality.

Resources

Equipment	Tools	Materials
● Computer ● Sensors ● Arduino boards ● Node MCU boards ● Raspberry pi	● XAMMP ● DBMS, MYSQL ● Web browser ● Text editor (Sublime text, ● Notepad++)	● Electricity ● Internet

Advance Preparation:

Before delivering this learning outcome, you are recommended to:

- Avail computer Lab with computers working in a good condition.

Indicative content 1.1: Analysis of IoT Web Application Requirements.

Duration: 5 hrs

Theoretical Activity 1.1.1: Description of IoT Web Application System

Notes to the trainer:

- Trainer may use small groups to describe IoT Web Application system requirement
- The use of images, videos, and illustrations as didactic materials is required.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:

i. Define IoT Web Application
ii. What are IoT Web Application requirement
iii. Identify the source of data
iv. Describe IoT system architecture
v. Describe code management
vi. Differentiate functional requirements from non- functional requirements

Step 2: Ask trainees to Write their answers on paper, blackboard, flipchart or white board

Step 3: Ask trainees to present their findings to the trainer.

Step 4: Provide expert view for more clarification

Step 5: Ask trainees to read the key readings 1.1.1. in trainee manual and ask question if necessary.

Points to Remember

- IoT Web Application is a web-based software platform designed to interact with and manage IoT devices. These applications collect, process, and display data from various connected devices and sensors, enabling users to monitor and control these devices remotely.

- There is different source of data in IoT web application which are: IoT Devices/Sensors, External APIs etc.
- Code Management in the context of IoT web applications involves practices and tools to manage the software development lifecycle efficiently that include continuous delivery, code minimization and optimisation
- The functional requirements define what the system should do, while non-functional requirements define how well the system performs those functions.



Theoretical Activity 1.1.2: Description on data collection



Notes to the trainer:

- Trainer may use small groups to describe data collection
- The use of images, videos, and illustrations as didactic materials is required.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:

i. What is data collection?
ii. What are the required devices use while developing and displaying data?

Step 2: Ask trainees to Write their answers on paper, blackboard, flipchart or white board

Step 3: Ask trainees to present their findings to the trainer.

Step 4: Provide expert view for more clarification

Step 5: Ask trainees to read the key readings 1.1.2. in trainee manual and ask question if necessary



Points to Remember

- Data Collection Determine what data the IoT application needs to collect, monitor, or control. This can include sensor data (temperature, humidity, motion, etc.), user input, or external data sources.
- The required devices use while developing and displaying data such as: Sensors and IoT Devices, Microcontrollers and Microprocessors, Data Storage Solutions, Data Processing Units, Networking Equipment, Display Devices, Development Tools.



Practical Activity 1.1.3: Analysing IoT web application requirements



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should select required devices and collect data.
- The use of videos as didactic materials is required.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to a computer lab to analyse IoT web application requirements,.

Step 2: Explain the task and provide clear work instruction.

Step 4: Demonstrate how to analyse IoT web application requirements. While demonstrating, explain the steps to follow.

Step 5: Asks trainees to analyse IoT web application requirements and monitor the activity.

Step 6: Verify whether the selection the required devices and collection of data used in IoT web application are properly performed.

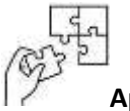
Step 7: Ask trainees to read key reading 1.1.3.



Points to Remember

- Analysing the requirements for an IoT (Internet of Things) web application involves a systematic approach to gather, evaluate, and document the necessary components to ensure the application functions effectively. Here are the steps to follow for analysing the requirements:

1. Identify and Analyze IoT Devices and Sensors
2. Data Management Requirements
3. Define User Requirements
4. Define Functional Requirements
5. Non-Functional Requirements
6. Integration with Third-Party Services
7. Security and Privacy Requirements



Application of learning 1.1.

XYZ Company want to develop an IoT web application for managing a smart building. This system will enable building managers to monitor and control energy usage, Lighting, HVAC (Heating, Ventilation, Air conditioning) through a web-based interface. you are requested to analyse the requirements by collecting data and Selecting the required devices for developing and displaying.

Checklist:

Criteria	**Indicator**	**Observation**	
		Yes	**No**
IoT web application system requirements are properly analysed	Identify system requirement is performed		
	Functional requirements is indicated		
	Non-functional requirements are gathered		
	Code management is perfomed		

Indicative content 1.2: Prepare PHP Environment

Duration: 5 hrs

Theoretical Activity 1.2.1: Description of PHP key terms

Notes to the trainer:

- Trainer may use small groups to describe PHP key terms.
- The use of images, videos, and illustrations as didactic materials is required.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:

i. What do you understand by the following terms: PHP, Interpreter, Compiler , Open source, Web server, browser, Text editor, Integrated Development E nvironment (IDE)?

ii. What is the purpose and characteristics of PHP?

iii. Describe the types of Database

Step 2: Ask trainees to Write their answers on paper, blackboard, flipchart or white board

Step 3: Ask trainees to present their findings to the trainer.

Step 4: Provide expert view for more clarification

Step 5: Ask trainees to read the key readings 1.2.1. in trainee manual and ask question if necessary.

Points to Remember

- PHP, which stands for "Hypertext Pre-processor," is a widely used open-source scripting language primarily designed for web development. There are purposes of PHP such as: PHP is designed primarily for server-side scripting, enabling developers to create dynamic and interactive web applications and characteristics are: Easy to Learn, Flexibility and Integration.
- There are different types of database such as: Relational Databases, NoSQL Databases, In-Memory Databases and Cloud Databases etc…



Practical Activity 1.2.2: Preparing PHP environment



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should install and configure of PHP environment.
- The use of videos as didactic materials is required.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to the computer lab to prepare PHP environmental tools:

Step 2: Explain the task and provide clear work instruction.

Step 4: Demonstrate how to prepare PHP environments. While demonstrating, explain the steps to follow.

Step 5: Asks trainees to prepare PHP environments and monitor the procedures.

Step 6: Verify whether Apache, mysql, PHP, Browser, Text Editor, and IDE are clearly installed.

Step 7: Ask trainees to read key reading 1.2.2.

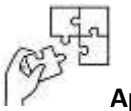


Points to Remember

These are the steps to follow when prepare PHP environment tools in IoT web application Development.

Step 1: Install a Web Server
Step2: Install PHP

Step3: Install a Database Server (Optional)
Step4: Configure the Environment
Step5: Install a Code Editor
Steps6.to install browser
Step 7: Test Your Environment



Application of learning 1.2.

XYZ Company wants to set up the server-side components necessary to connect IoT devices with a central server and ensure data is stored and processed effectively, you are requested to prepare a PHP-based web environment for an IoT application.

Checklist:

criteria	**Indicator**	**Observation**	
		Yes	**No**
PHP environments are properly prepared	a Web Server Installed		
	Database Server is correctly installed		
	PHP is installed		
	PHP Environment Configured		
	a Code Editor is installed		
	Browser is installed		
	PHP Environment is Tested		

Indicative content 1.3: Identification of PHP Concepts

Duration: 10hrs

Theoretical Activity 1.3.1: Description of PHP Concepts

Notes to the trainer:

- Trainer may use small groups to describe PHP concepts
- Have videos as didactic materials is required.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:

i. Define the following terms used in PHP and give an example per each:

a) Tags
b) Super global variable
c) Operators
d) Data types
e) Variable scope
f) Constant
g) Comment
h) String concatenation

ii. What are the types of variables?

iii. What are the naming rule of variable?

Step 2: Ask trainees to Write their answers on paper, blackboard, flipchart or white board

Step 3: Ask trainees to present their findings to the trainer.

Step 4: Provide expert view for more clarification

Step 5: Verify whether,Tags,Variable,Naming rules,Types Scopes,Super Global variables ,Operators,Data types,Variable scope,Constants,Comment ,String concatenation are identified.

Step 6: Ask trainees to read the key readings 1.3.1. in trainee manual and ask question if necessary

Points to Remember

In PHP programming they are different naming rules of variable which are: Variable na mes must start with a dollar sign,Variable names must not start with numbers,Variable names include letters, numbers, and underscores, Variable names are case-sensitive.

Theoretical Activity 1.3.2: Description of fundamental programming concepts

Notes to the trainer:

- Trainer may use small groups to describe PHP concepts
- Have videos as didactic materials is required.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:

i..What do you understand by the following terms?

a) Control structure
b) Array
c) Functions
d) File handling
e) Data validation

ii. What are the components of Control structure.
iii. What are the Types of Arrays?
iv. What are the Types of Functions.
v. What are file operations in PHP programming?

Step 2: Ask trainees to Write their answers on paper, blackboard, flip-chart or white board

Step 3: Ask trainees to present their findings to the trainer.

Step 4: Provide expert view for more clarification

Step 5: Verify whether control structure, arrays, functions,file handling,data validation are properly described

Step 6: Ask trainees to read the key readings 1.3.1. in trainee manual and ask question if necessary



Points to Remember

- Control structure in PHP are essential for directing the flow of execution in a program. Such as conditional structure, Looping Structures, Control Flow Statements, Switch statement, jump statement.
- Arrays in PHP are a powerful and versatile data structure that allows you to store multiple values in a single variable array has 2 types single and multi dimensional array
- Functions in PHP are blocks of code that perform a specific task and can be reused throughout your program such as Defining a Function, Function with Parameters, Returning Values, Default Parameter Values etc.
- File handling in PHP allows you to read from and write to files on the server. PHP provides a range of functions to perform various file operations, such as creating, opening, reading, writing, and deleting files



Practical Activity 1.3.2: applying programming fundamentals



Notes to the trainer

- This activity should take place in computer lab where the trainees will be applying programming fundamentals in PHP.
- Have videos as didactic materials is required.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to the computer lab to apply programming fundamentals concepts in PHP

Step 2: Explain the task and provide clear work instruction.

Step 3: Demonstrate how to apply programming fundamentals concepts in PHP. While demonstrating, explain the steps to follow.

Step 4: Asks trainees to apply programming fundamentals concepts in PHP and monitor the procedures.

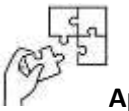
Step 5: Verify whether control structure, arrays, functions,file handling, data validation are properly applied.

Step 6: Ask trainees to read key reading 1.3.2.



Points to Remember

- While applying PHP programming fundamental pass to the following steps:
  Step 1: Set Up Your Development Environment
  Step 2: Learn Basic Syntax
  Step 3: Control Structures
  Step 4: Functions
  Step 5: Arrays
  Step 6: File Handling
  Step 7: Data Validation and Sanitization



Application of learning 1.3.

You have been hired to develop a small system for collecting and managing customer feedback for a local business. The system will allow users to submit their feedback, store it in a file, display all feedback entries, and delete outdated or inappropriate feedback. Additionally, the system will validate the inputs to ensure only appropriate feedback is accepted.

Checklist:

criteria (Based on	Indicator	Observation	
		Yes	No
PHP fundamentals is properly performed	Set Up Your Development Environment is indicated		
	Syntax is used		
	Control Structures is perfomed		
	Functions is perfomed		
	Arrays perfomed		
	File Handling perfomed		
	Data Validation and Sanitization		

Indicative content 1.4: Develop Application Programming Interface CRUD Endpoints

Duration: 10 hrs

Theoretical Activity 1.4.1: Description of API CRUD Endpoints

Notes to the trainer:

- Trainer may use small groups to describe API concept.
- The use of images, videos, and illustrations as didactic materials is required.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:

i. Define Application Programming Interface (API) and Endpoints
ii. Explain the types of Representational State Transfer REST) and Simple Object Access Protocol (SOAP)
iii. What are the API working Principles?
iv. Describe the API Request methods
v. Describe the endpoint used in API?
vi. What is the application of data format JSON?

Step 2: Ask trainees to Write their answers on paper, blackboard, flip-chart or white board

Step 3: Ask trainees to present their findings to the trainer.

Step 4: Provide expert view for more clarification

Step 5: Ask trainees to read the key readings 1.4.1. in trainee manual and ask question if necessary

Points to Remember

- An API is a set of rules and protocols for building and interacting with software applications. It defines the methods and data formats that applications can use to communicate with each other, enabling different software systems to interact and share data.

- In IoT Web Application Development using PHP, the API Request Methods used are GET, POST, PUT, DELETE and PATCH.
- The REST (Representational State Transfer) is an architectural style for designing networked applications, SOAP (Simple Object Access Protocol): is a protocol used for exchanging structured information in web services and JSON (JavaScript Object Notation) is a lightweight data interchange format that is easy for humans to read and write and easy for machines to parse and generate.



Practical Activity 1.4.2: Applying of data format (JSON)



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should Apply JSON format.
- The use of videos as didactic materials is required.
- Avail computers installed with PHP environment and text editor



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to the computer lab to Apply JSON format in PHP.

Step 2: Explain the task and provide clear work instruction.

Step 4: Demonstrate how to apply data format JSON in PHP. While demonstrating, explain the steps to follow.

Step 5: Asks trainees to Apply data format JSON in PHP. and monitor the procedures.

Step 6: Verify whether the data format JSON is properly performed**.**

Step 7: Ask trainees to read key reading 1.4.2.

Points to Remember

- The steps to follow when Applying data format JSON in IoT web application development using PHP are:
  - Step 1: Define the Data Structure
  - Step 2: Set Up the PHP Environment
  - Step 3: Create the Database Schema
  - Step 4: Develop API Endpoints
  - Step5: Handle Incoming JSON Data
  - Step6: Validate the Data
  - Step7: Process the Data
  - Step8: Send JSON Responses
  - Step9: Error Handling
  - Step10: Testing the API
  - Step11: documentation



Practical Activity 1.4.3: Executing CREATE Query



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should Create a query using CRUD endpoints.
- The use of videos as didactic materials is required.
- Avail computers installed with PHP environment and text editor



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to the computer lab to Execute CREATE Query.

Step 2: Explain the task and provide clear work instruction.

Step 4: Demonstrate how to Execute CREATE Query. While demonstrating, explain the steps to follow.

Step 5: Asks trainees to Execute CREATE Query and monitor the procedures.

Step 6: Verify whether the register IoT device, validate data from IoT device,store data in database is properly performed**.**

Step 7: Ask trainees to read key reading 1.4.3.



Points to Remember

- The steps to follow when creating a query in IoT web application development using PHP are:

  Step 1: Set Up Your Environment

  Step 2: Create a Database and Table

  Step 3: Connect to the Database in PHP

  Step 4: Create the Query to Insert Data and execute

  Step5: Prepare the Statement

  Step6: Bind Parameters

  Step7: Handle the Response

  Step8: Close the Connection

  Step9: Error Handling

  Step10: Testing



Practical Activity 1.4.4: Execution of READ Query



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should execute READ a query using PHP environment.
- The use of videos as didactic materials is required.
- Avail computers installed with PHP environment and text editor

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:
As web developer (IoT), you are asked to go to the computer lab to Execute READ Query.
Step 2: Explain the task and provide clear work instruction.
Step 4: Demonstrate how to Execute RAED Query. While demonstrating, explain the steps to follow.
Step 5: Asks trainees to Execute READ Query and monitor the procedures.
Step 6: Verify whether the retrieve device information,fetch data from external application programming interface,check device status,access alert record is properly performed**.**
Step 7: Ask trainees to read key reading 1.4.4.

Points to Remember

In IoT web application development using PHP here are the procedures of execute the Read Query

Step 1: Set Up Your Environment

Step 2: Create a Database and Table

Step 3: Connect to the Database in PHP

Step 4: Create the Read Query

Step 5: Test the Read Query

Step 6: Handling Errors and Enhancements

Practical Activity 1.4.5: Executing UPDATE Query

Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should execute UPDATE a query using PHP environment.
- The use of videos as didactic materials is required.

- Avail computers installed with PHP environment and text editor

 **Key steps:**

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:
As web developer (IoT), you are asked to go to the computer lab to Execute UPDATE Query.
Step 2: Explain the task and provide clear work instruction.
Step 4: Demonstrate how to Execute UPDATE Query. While demonstrating, explain the steps to follow.
Step 5: Asks trainees to Execute UPDATE Query and monitor the procedures.
Step 6: Verify whether update device information, modify sensor settings, update device firmware, update access permission update alert and notification settings is properly performed**.**
Step 7: Ask trainees to read key reading 1.4.5.

 **Points to Remember**

- In IoT web application development using PHP here are the procedures of execute the Update Query

  Setp 1: Set Up Your Environment

  Setp 2: Create a Database and Table

  Setp 3: Connect to the Database in PHP

  Setp 4: Prepare the UPDATE SQL Statement

  Setp 5: prepare the statement

  Setp 6: Bind Parameters

  Setp 7: Execute the Query

  Setp 8: Test the Update Query

  Setp 9: Close the Connection



Practical Activity 1.4.6: Execution of Delete Query and error handling



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should execute Delete Query and error handling using PHP environment.
- The use of videos as didactic materials is required.
- Avail computers installed with PHP environment and text editor



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to the computer lab to Execute Delete Query and error handling.

Step 2: Explain the task and provide clear work instruction.

Step 4: Demonstrate how to Execute Delete Query and error handling. While demonstrating, explain the steps to follow.

Step 5: Asks trainees to Execute Delete Query and error handling and monitor the procedures.

Step 6: Verify whether the remove sensor data, delete command history, purge inactive devices, delete alert configurations and error handling is properly performed**.**

Step 7: Ask trainees to read key reading 1.4.6.

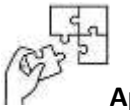


Points to Remember

- In IoT web application development using PHP here are the procedures of Delete Query and error handling.

  Step 1: Set Up Your Environment

Step 2: Create a Database and Table
Step 3: Connect to the Database in PHP
Step 4: Create the Delete Query
Step 5: Prepare the Statement
Step6: Bind Parameters
Step7: Execute the Query
Step8: Close the Connection
Step 9: Test the Delete Query
Step 10: Error Handling



Application of learning 1.4.

XYZ Company is developing an IoT Device Management System to monitor and manage connected devices remotely. They require an Application Programming Interface **(**API**)** to handle device registration, data retrieval, updates to device settings, and device removal. The API will follow REST principles and use JSON as the primary data format. The API must also handle errors efficiently.

As a developer, you are tasked with building this API using PHP and MySQL to perform CRUD (Create, Read, Update, Delete) operations on the IoT device database.

Checklist:

Criteria	Indicator	Observation	
		Yes	No
Develop Application Programming Interface CRUD Endpoints	CREATE query is Executed		
	READ Query is Execute		
	UPDATE Query is Executed		
	DELETE Query is Executed		
	Error Handling is perfomed		

Indicative content 1.5: Secure API Endpoints

Duration: 4 hrs

Theoretical Activity 1.5.1: Description of API Endpoints Secure

Notes to the trainer:

- Trainer may use small groups to Description of API Endpoints Secure.
- The use of images, videos, and illustrations as didactic materials is required.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:
  i. What is authentication
  ii. What are API security best practices?

Step 2: Ask trainees to Write their answers on paper, blackboard, flip-chart or white board

Step 3: Ask trainees to Present their findings to the trainer.

Step 4: Provide expert view for more clarification

Step 5: Ask trainees to read the key readings 1.5.1. in trainee manual and ask question if necessary

Points to Remember

- the best practice for API security are: implementing Authentication methods, use HTTPS, validate input, use rate limit, limiting data exposure, monitor and log API activities, keep API updated, Use Strong Authentication Methods.



Practical Activity 1.5.2: Implementing API authentication



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should implement API authentication.
- The use of videos as didactic materials is required.
- Avail computers installed with PHP environment and text editor



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to the computer lab to implement API Authentication.

Step 2: Explain the task and provide clear work instruction.

Step 3: Demonstrate how to implement API Authentication. While demonstrating, explain the steps to follow.

Step 4: Asks trainees to implement API Authentication and monitor the procedures.

Step 5: Verify whether the implementation of API Authentication is properly performed**.**

Step 6: Ask trainees to read key reading 1.5.2.



Points to Remember

- API authentication can be implemented in IoT web application development using the following steps:
  1. Define Authentication Method
  2. Set Up Database for User/Device Management
  3. Create Registration Endpoint

4. Create Login Endpoint
5. Implement Token Generation



Practical Activity 1.5.3: Applying API Security Best Practices



Notes to the trainer

- Trainer may use small groups to apply API security best practices.
- The use of images, videos, and illustrations as didactic materials is required.



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to the computer lab Applying API Security Best Practices

Step 2: Explain the task and provide clear work instruction.

Step 3: Demonstrate how to apply API Security Best Practices. While demonstrating, explain the steps to follow.

Step 4: Asks trainees to Apply API Security Best Practices and monitor the procedures.

Step 5: Verify whether the Application of API Security Best Practices is properly performed**.**

Step 6**:** Ask trainees to read key reading 1.5.3.



Points to Remember

Steps for API Security Best Practices in PHP

Step 1. Use HTTPS
Step 2. Authentication and Authorization
Step 3. Input Validation and Sanitization
Step 4. Rate Limiting

Step 5. Error Handling
Step 6. CORS (Cross-Origin Resource Sharing) Configuration
Step 7. Use API Keys
Step 8. Data Encryption
Step 9. Implement Logging and Monitoring
Step 10. Keep Software Updated



Practical Activity 1.5.4: Managing API security



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should manage API security.
- The use of videos as didactic materials is required.
- Avail computers installed with PHP environment and text editor



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to the computer lab to manage API security.

Step 2: Explain the task and provide clear work instruction.

Step 3: Demonstrate how to manage API security. While demonstrating, explain the steps to follow.

Step 4: Asks trainees to manage API security and monitor the procedures.

Step 5: Verify whether the Execution of managing API security is properly performed**.**

Step 7: Ask trainees to read key reading 1.5.4.

Points to Remember

The steps for managing API security in IoT web application development using PHP are:

1. Use HTTPS
2. Implement Authentication
3. Use Strong Password Policies
4. Rate Limiting
5. Input Validation and Sanitization
6. Use CORS (Cross-Origin Resource Sharing)
7. Implement Logging and Monitoring
8. Use Security Headers
9. Regular Security Audits
11. Implement Device Authentication
12. Use Firewalls and Security Groups

Application of learning 1.5.

XYZ Company wants to create API endpoint necessary to connect IoT devices with a central server and ensure data is stored and processed effectively, you are requested to manage API security for an IoT web application.

Checklist:

Criteria	Indicator	Observation	
		Yes	No
Secure API endpoints	Application Programming Interface authentication is Implemented		
	Application Programming Interface security Best practices is Applied		
	Application Programming Interface security is Managed		

Indicative content 1.6: Test Application Programming Interface Endpoint

Duration: 4 hrs

Theoretical Activity 1.6.1: Description of API testing concept

Notes to the trainer:

- Trainer may use small groups to describe API testing concept
- The use of images, videos, and illustrations as didactic materials is required.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following questions:

i. What are the API testing Tools?
ii. What are API testing best practices?
iii. Give the benefits of Application Programming Interface testing
iv. What are types of application programming interface endpoints testing

Step 2: Ask trainees to Write their answers on paper, blackboard, flip-chart or white board

Step 3: Ask trainees to Present their findings to the trainer.

Step 4: Provide expert view for more clarification

Step 5: Ask trainees to read the key readings 1.6.1. in trainee manual and ask question if necessary.

Points to Remember

- There are numerous tools available for API testing, each with its own strengths. Here are some popular ones such as postman,soapUI, RESTassued, JMeter,swagger etc.



Practical Activity 1.6.2: Testing API accessibility



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should test API accessibility.
- The use of videos as didactic materials is required.
- Avail computers installed with PHP environment and text editor



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to the computer lab to test API accessibility.

Step 2: Explain the task and provide clear work instruction.

Step 4: Demonstrate how to test API accessibility. While demonstrating, explain the steps to follow.

Step 5: Asks trainees to test API accessibility and monitor the procedures.

Step 6: Verify whether the unit testing, integration testing, end to end testing, load testing is properly performed**.**

Step 7: Ask trainees to read key reading 1.6.2.



Points to Remember

Steps to test API accessibility

Step 1: Set Up Your Testing Environment
Step 2: Define the API Endpoints
Step 3: Write Test Cases

Step 4: Run the Tests
Step 5: Validate Responses
Step 6: Test Edge Cases
Step 7: Automate Tests (Optional)



Application of learning 1.6.

XYZ Company wants to create API endpoint necessary to connect IoT devices with a central server and ensure data is stored and processed effectively, you are requested to test API accessibility for an IoT web application.

Checklist:

criteria (	**Indicator**	**Observation**	
		Yes	**No**
Test Application Programming Interface endpoint	Testing tools is used		
	API testing best practices is applied		
	Testing steps is indicated		
	Application Programming Interface endpoints testing is used		
	Unit testing is perfomed		
	Integration testing is perfomed		
	End to end testing is perfomed		
	Load testing is perfomed		

Indicative content 1.7: Documentation of the API

Duration: 2 hrs

Theoretical Activity 1.7.1: Identification of documentation tools for developed

Notes to the trainer:

- Trainer may use small groups to identify tools for developing API.
- The use of images, videos, and illustrations as didactic materials is required.

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity by asking trainees to answer the following question s:

a) Define API versioning
b) What are the documentation tools for developed API?
c) What is API specification (information) documentation?

Step 2: Ask trainees to Write their answers on paper, blackboard, flip-chart or white board

Step 3: Ask trainees to Present their findings to the trainer.

Step 4: Provide expert view for more clarification

Step 5: Ask trainees to read the key readings 1.7.1. in trainee manual and ask question if necessary

Points to Remember

- API Versioning is the practice of managing changes to an API in a way that allows developers to introduce new features, fix bugs, or make improvements without disrupting existing clients that rely on the current API. Versioning ensures backward compatibility and provides a clear path for clients to adopt new functionality at their own pace
- Documentation tools used in development of API such as Swagger (OpenAPI), Postman, Apiary etc.

- The API Specification (Information) Documentation include Endpoint (GET, POST, PUT, DELETE), Request Parameters, Response Structures.



Practical Activity 1.7.2: Documentation of API endpoint



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should document API endpoint.
- The use of videos as didactic materials is required.
- Avail computers installed with PHP environment and text editor



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:

As web developer (IoT), you are asked to go to the computer lab to provide steps to document API Endpoint.
Step 2: Explain the task and provide clear work instruction.
Step 4: Demonstrate how to document API Endpoint. While demonstrating, explain the steps to follow.
Step 5: Asks trainees to document API Endpoint and monitor the procedures.
Step 6: Verify whether API Endpoint (Include authentication information, Include Request parameters, Include Response structure, Include Error codes and error response formats, add usage guidelines) is properly performed**.**
Step 7**:** Ask trainees to read key reading 1.7.2.

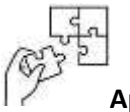


Points to Remember

steps to document API Endpoint.

Step 1: Choose a Documentation Format
Step 2: Provide an Overview of the API
Step 3: Include Authentication Information
Step 4: Document Request Parameters

Step 5: Document Response Structure
Step 6: Document Error Codes and Error Response Formats
Step 7: Add Usage Guidelines
Step 8: Review and Update Regularly
Step 9: Publish the Documentation



Application of learning 1.7.

XYZ Company wants to create API endpoint necessary to connect IoT devices with a central server and ensure data is stored and processed effectively, you are requested to document API endpoint for an IoT web application.

Checklist:

Criteria	Indicator	Observation	
		Yes	No
Documentation of the developed API	documentation tools is used		
	Document API specifications is indicated		
	authentication information is included		
	Request parameters is included		
	Response structure is included		
	Error codes and error response formats is included		
	usage guidelines is added		



Learning outcome 1 end assessment

Theoretical assessment

I.Multiple Choice Questions

Question 1:What is the primary purpose of an API?

A) To store data

B) To enable communication between different software systems

C) To create user interfaces

D) To manage databases

Answer: B) To enable communication between different software systems

Question 2:Which HTTP method is typically used to retrieve data from an API?

A) POST

B) PUT

C) GET

D) DELETE

Answer: C) GET

Question 3:What PHP function is commonly used to encode data into JSON format?

A) json_decode()

B) json_encode()

C) json_format()

D) json_stringify()

Answer: B) json_encode()

Question 4:Which of the following is a common way to secure an API?

A) Using plain text for passwords

B) Implementing API keys or tokens

C) Allowing all IP addresses to access the API

D) Disabling authentication

Answer: B) Implementing API keys or tokens

Question 5: What is the purpose of using a PHP for API development?

A) To avoid using PHP

B) To provide a structured way to build applications and APIs

C) To create HTML pages

D) To manage databases directly

Answer: B) To provide a structured way to build applications and APIs

Question 6:Which status code indicates that a request was successful and the server returned the requested data?

A) 404

B) 500

C) 200

D) 403

Answer: C) 200

Question 7:

What is the purpose of using middleware in a PHP API application?

A) To handle database connections

B) To process requests before reaching the main application logic

C) To render HTML pages

D) To manage user sessions

Answer: B) To process requests before reaching the main application logic

Question 8:Which of the following is a common tool for testing APIs?

A) MySQL

B) Postman

C) HTML

D) CSS

Answer: B) Postman

Question 9:When developing an API, what does CORS stand for?

A) Cross-Origin Resource Sharing

B) Common Object Resource System

C) Centralized Open Resource Service

D) Cross-Origin Resource Security

Answer: A) Cross-Origin Resource Sharing

Question 10:What is the primary benefit of using RESTful principles in API design?

A) It allows for real-time communication.

B) It provides a standardized way to interact with resources.

C) It eliminates the need for authentication.

D) It allows for complex data queries.

Answer: B) It provides a standardized way to interact with resources.

II. . Read the following statement related to developing APIs using PHP, and answer by True if the statement is correct or False if the statement is wrong

Question 1:APIs can only be developed using JavaScript and cannot be created using PHP.

Answer: False

Question 2:The HTTP method POST is typically used to create new resources in a RESTful API.

Answer: True

Question 3:JSON (JavaScript Object Notation) is the only format that can be used for data exchange in APIs.

Answer: False

Question 4:Using HTTPS instead of HTTP is important for securing API communication.

Answer: True

Question 5:API keys are a form of authentication that can be included in the request headers.

Answer: True

Question 6:It is unnecessary to validate user input when developing an API because the API is only accessed by trusted clients.

Answer: False

Question 7:The status code 404 indicates that the requested resource was not found on the server.

Answer: True

Question 8:Middleware in a PHP API can be used to handle tasks such as authentication, logging, and request modification.

Answer: True

Question 9:A RESTful API must always use the same URL structure for all endpoints.

Answer: False

Question 10:It is a good practice to document your API endpoints for better usability and integration by other developers.

Answer: True

Open questions

Question 1: Define IoT Web Application

Question 2: Identify Source of Data

Question 3: What is an API, and why is it important in web development?

Question 4: Explain the difference between RESTful APIs and SOAP APIs.

Question 5: Describe the different HTTP methods (GET, POST, PUT, DELETE) and their typical uses in RESTful APIs.

Question 6: How can PHP be used to create an API? What are the advantages of using PHP for API development?

Question 7: Discuss common security practices for API development, such as authentication and data validation.

Answers

Question 1: Define IoT Web Application
An IoT Web Application is a platform that facilitates the interaction, monitoring, and management of IoT devices through a web interface. It processes data collected from IoT devices, such as sensors and actuators, and presents it to users in a meaningful way.

Question 2: Identify Source of Data
In an IoT Web Application, data sources include sensors that capture real-time data such as temperature or motion, databases that store historical or configuration data, APIs that fetch external data like weather or stock prices, actuators that provide feedback from devices, and user input collected through forms or dashboards.

Question 3: What is an API, and why is it important in web development?
An API, or Application Programming Interface, is a set of rules and protocols that enable

communication between different software applications. It is essential in web development as it facilitates integration by allowing developers to connect various systems like web apps, databases, and external services.

Question 4: Explain the difference between RESTful APIs and SOAP APIs

RESTful APIs are lightweight and web-friendly, using HTTP protocols and supporting multiple data formats like JSON and XML. They are flexible, easy to implement, and ideal for web and mobile applications requiring scalability. In contrast, SOAP APIs are protocol-based, relying on XML for communication. While SOAP follows strict standards and is suitable for complex, enterprise-level applications, it is less flexible and slower compared to REST due to the overhead of XML parsing.

Question 5: Describe the different HTTP methods (GET, POST, PUT, DELETE) and their typical uses in RESTful APIs

HTTP methods define the actions performed on resources in RESTful APIs. The GET method retrieves data from the server, commonly used to fetch a list or specific resource. POST is used to create new resources, such as adding data to a server. PUT updates or completely replaces an existing resource, while DELETE removes a resource from the server. These methods collectively enable CRUD (Create, Read, Update, Delete) operations in web development.

Question 6: How can PHP be used to create an API? What are the advantages of using PHP for API development?

PHP can be used to create an API by setting up a server environment (e.g., Apache, MySQL, PHP), handling HTTP requests, processing data, and connecting to databases using extensions like PDO or MySQLi. Developers can structure endpoints using REST principles and return responses in JSON format with PHP's json_encode() function. PHP is advantageous for API development due to its simplicity, cost-effectiveness, wide community support, rich libraries, and seamless integration with databases and front-end frameworks.

Question 7: Discuss common security practices for API development, such as authentication and data validation

API security is crucial in web development, and practices like authentication and data validation play a vital role. Authentication methods such as API keys, OAuth, and JSON Web Tokens (JWT) ensure that only authorized users can access APIs. Data validation is essential to prevent injection attacks and involves validating inputs and using prepared statements for secure handling.

Advantages of using PHP for API development:

- **Ease of use**: PHP is a simple and widely-known language, making it accessible for developers.

- **Large ecosystem**: PHP has a wide array of libraries and frameworks (like Laravel, Slim) that speed up API development.
- **Built-in functions**: PHP provides robust functions for handling requests, JSON encoding/decoding, and interacting with databases.
- **Performance**: PHP is optimized for web-based applications, making it efficient for API development.
- **Cross-platform compatibility**: PHP works well across different server environments (Linux, Windows).

Question 5: Discuss common security practices for API development, such as authentication and data validation.

Common security practices in API development include:

- **Authentication**:
  - Implement secure user authentication methods like OAuth 2.0, JWT (JSON Web Tokens), or API keys to ensure that only authorized clients can access the API.
- **Data Validation**:
  - Always validate incoming data (e.g., request payloads) to avoid injection attacks (like SQL injection or XSS). Input should be sanitized and validated both on the client and server sides.
- **Rate Limiting**:
  - Restrict the number of requests a client can make in a given period to prevent Denial-of-Service (DoS) attacks.
- **Encryption (HTTPS)**:
  - Always use HTTPS to encrypt data in transit between the client and server. This ensures that sensitive information (like passwords) is not exposed.
- **CORS (Cross-Origin Resource Sharing)**:
  - Implement CORS policies to control which domains can access the API, preventing unauthorized clients from making requests.
- **Error Handling**:
  - Provide detailed error messages for debugging, but avoid exposing sensitive information (e.g., database structure, server details) in production error messages.

Practicatical assessment

XYZ company want to creating endpoints that allow clients (like web applications or IoT devices) to interact with their server. As an IoT web application developer you are requested to create a simple RESTful API for managing tasks which will allow users to create, read, update, and delete.

Checklist:

Criteria (	Indicator	Observation	
		Yes	No
Application programming interface (API) using PHP IS properly developed	IoT Web application requirements is analysed		
	PHP environment is prepared		
	PHP concepts are identified		
	Application Programming Interface CRUD Endpoints are developed		
	API endpoints are secured		
	API endpoints are tested		



Further information to the trainer

Prabowo, Muhamad Cahyo Ardi, et al. "Development of an IoT-Based Egg Incubator with PID Control System and Web Application." JOIV: International Journal on Informatics Visualization 8.1 (2024): 465-472.

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Mohammad El-Basioni, Basma M., and Sherine M. Abd El-Kader. "Designing and modeling an IoT-based software system for land suitability assessment use case." Environmental Monitoring and Assessment 196.4 (2024): 380.

https://www.geeksforgeeks.org/introduction-to-internet-of-things-iot-set-1/

Azkiya, Khoirul, Muhamad Irsan, and Muhammad Faris Fathoni. "Implementation of App Engine and Cloud Storage as REST API on Smart Farm Application." Sinkron: jurnal dan penelitian teknik informatika 8.2 (2024): 902-910


CSS
HTML

Indicative Contents
2.1 Identification of UI Requirements
2.2 Use HTML Tags
2.3 Application of CSS

Key Competencies for Learning Outcome 2: Develop User Interface

Knowledge	Skills	Attitudes
• Identification of user • interface requirements • Description of HTML tag • Description of CSS	• Conducting system Stakeholder to gather requirement • Creating User Personas from Data • Developing User Interface • Understanding HTML and PHP Basics • Installing and Configuring a HTML and PHP Environment • Setting Up a Local Development Environment • Connecting HTML and PHP to a Database • Understanding HTML Structure • Writing and Understanding HTML Tags • Understanding CSS Syntax • Applying Styling Properties	• Being Open-Mindedness from stakeholders • Having communication skills to clearly articulate CSS concepts • Having User-Centric Focus and behaviors of the end users. • Having commitment for Learning • Having Attention to Detail for installation and configuration steps • Having Curiosity to Maintain and understanding how PHP interacts with databases • Having Logical Thinking to understand how HTML tags fit into the overall document structure

Duration: 25hrs

Learning outcome 2 objectives:

By the end of the learning outcome, the trainees will be able to:

1. Identify properly user interface requirement Used to develop user interface in IOT web application
2. Describe clearly HTML tag Used to develop user interface in IOT web application
3. Describe properly css Used to develop user interface in IOT web application
4. Installing and Configuring a HTML and PHP Environment Used to develop user interface in IOT web application
5. Applying Styling Properties Used to develop user interface in IOT web application
6. Understanding HTML and PHP Basics Used to develop user interface in IOT web application
7. Understanding CSS Syntax Used to develop user interface in IOT web application

Resources

Equipment	Tools	Materials
● Computer ● Sensors ● Arduino boards ● Node MCU boards ● Raspberry pi	● XAMMP ● DBMS, MYSQL, postman ● Web browser ● Text editor (Sublime text, ● Notepad++)	● Electricity ● Internet ●

Advance Preparation:

Before delivering this learning outcome, you are recommended to:

- Avail equipment, tools, material and software used in Iot web application for teaching develop user interface
- Avail working environment

Indicative content 2.1: Identification of UI Requirements

Duration: 5hrs

Theoretical Activity 2.1.1: Identification of User Interface Requirement

Notes to the trainer:

- ✓ Trainer may use small group to identify user interface requirement through Real-World Examples
- ✓ The use of images or screenshots of real-world user interfaces to demonstrate clear navigation, consistent design, and familiar UI patterns

Key steps:

While delivering this activity, pass through the following steps:

Step1: Introduce the activity by asking trainees to answer the following questions:

a. What is user personas
b. What are Review System requirements?
c. Provide List key user services in develop user interface
d. Describe user experience (UX) best practices
e. What are the Features of a good user interface?

Step2: Write your answers on paper, blackboard, flip-chart or white board

Step3: Present your findings to the trainer or your classmates

Step4: Ask question for clarification if any.

Step5: Read the key readings 2.1.1.

Points to Remember

- There are user experience (UX) best practices which are: Define user interface, Clear, Intuitive Navigation, Consistent Design, Use Familiar UI Patterns, Choose an attractive colour scheme.



Practical Activity 2.1.2: Draw Wireframe



Notes to the trainer

- This activity should be carried out in a computer lab where the trainees should draw wiframe
- The use of videos as didactic materials is required.
- Avail computers installed with PHP environment and text editor



Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the topic and ask trainees to do the task described below:
As web developer (IoT), you are asked to go to the computer lab to draw wireframe
Step 2: Explain the task and provide clear work instruction.
Step 3: Demonstrate how to draw wireframe. While demonstrating, explain the steps to follow.
Step 4: Asks trainees to draw wireframe and monitor the procedures.
Step 5: Verify whether wireframe is properly drawn**.**
Step 6: Ask trainees to read key reading 2.1.2.



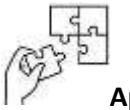
Points to Remember

Here are the steps to draw a wireframe for developing a user interface in PHP

Step 1: Define the Purpose and Scope
Step 2: Gather Requirements
Step 3: Choose Wire framing Tools
Step 4: Create a Site Map
Step 5: Start with Low-Fidelity Wireframes
Step 6: Focus on Key UI Elements
Step 7: Add Annotations

Step 8: Review and Iterate

Step 9: Create



Application of learning 2.1.

After understanding Identification of UI requirements: You are tasked with designing the user interface for a library management application. The app will be used by librarians, library members, and admin staff to manage book lending, returns, inventory, and user accounts.

Checklist

Criteria	Indicator	Observation	
		Yes	No
UI requirements	All system requirements are reviewed and met		
	Key services are listed		
	User personas is defined		
	The user interface is defined		
	Clear and intuitive navigation,consistent design,user familiar UIpatterns is indicated		
	Attractive colour scheme is choosen		
	A wireframe is drawned		

Indicative content 2.2: Use HTML Tags

Duration: 10hrs

Theoretical Activity 2.2.1: Description of HTML tag

Notes to the trainer:

- ✓ While delivering this content, a small group can be used for introducing the use of HTML tag.
- ✓ The use of visual aids like diagrams or slides to explain the hierarchy of HTML tags.

Key steps:

While delivering this activity, pass through the following steps:

1. Involve trainees in group formulation
2. Introduce the activity and ask learners to respond to the following questions:
    i. Provide the description of:
        a) HTML
        b) Tag
        c) Version of HTML?
        d) Tag attribute
    ii. What are HTML tag categories.
3. Ask trainees to present their findings to the whole class.
4. Provides expert view and clarifies ideas by using didactic materials.
5. Address any questions or concerns.
6. Ask trainee to read the key readings on activity 2.2.2

Points to Remember

- HTML, or Hypertext Mark-up Language, is the standard mark-up language used to create and design the structure of web pages. An HTML tag is a special code used in HTML to define the structure and content of a webpage. HTML tags are enclosed in angle brackets (< >) and usually come in pairs (an opening tag and a closing tag). And the can be used to create a website
- The main attribute is class, id, style, href etc.

- HTML categories are (HTML Structural tags, formatting tags, Table tags, Form tags, heading tags, List tags, Media tags, Code tags, HTML frame tags, HTML Comment, Grouping tags (div, span) Hyperlink tag, Semantic tags).



Practical Activity 2.2.2: Using HTML Tag



Notes to the trainer

- This activity should take place in a computer lab where trainees should structure the web using appropriate HTML tag
- Avail computers
- Avail text editor
- Avail browser



Key steps:

While delivering this activity, pass through the following steps:

Step1: Introduce the topic and ask trainees do the task described below:

> As an IoT web application developer you are requested to go the computer lab , to use HTML tags.

Step2: Explain the task and provide clear work instruction.

Step3: Demonstrate how to to use HTML tags.While demonstrating, explain the steps to follow.

Step4: Ask learners to to use HTML tags and monitor the procedures.

Step5: Verify whether the HTML Structural tags, formatting tags, Table tags, Form tags, heading tags, List tags, Media tags, Code tags, HTML frame tags, HTML Comment, Grouping tags (div, span) Hyperlink tag, Semantic tagsis correctly structured.

Step6: Ask trainees to read key reading 2.2.2.



Points to Remember

Step 1: Set Up Your HTML Document
Step 2: Use Structural Tags
Step 3: Add Additional Structural Tags
Step 4: Final HTML Structure
Step 5: Style Your Page
Step 6: Test Your Web Page



Application of learning 2.2.

You are part of a student organization at your schools and have been tasked with creating a webpage for an upcoming student fair. The page will showcase event details, a registration form for student groups, a schedule of activities, a gallery of previous events, and links to social media pages.

Checklist

Criteria	Indicator	Observation	
		Yes	No
The webpage structure using appropriate HTML tag	HTML structure Tags are used		
	Formatting tags is performed		
	Table tags is used		
	Form tags is indicated		
	Heading tags is performed		
	List tags is performed		
	Media tags is performed		
	Code tags is performed		

	HTML frame tags is performed		
	HTML Comment is performed		
	Grouping tags (div, span) is performed		
	Hyperlink tag is performed		
	Semantic tags is performed		

Indicative content 2.3: Applying CSS

Duration: 10hrs

Theoretical Activity 2.3.1: Description of CSS

Notes to the trainer:

- While delivering this content, a small group can be used for describing CSS.
- Avail computer connected to the internet
- Avail of computer installed with text editor

Key steps:

While delivering this activity, pass through the following steps:

Step1: Introduce the activity and request learners to respond to the following questions:

i. What do you understand about CSS?
ii. What are version of CSS.
iii. What are Types of CSS Styles
iv. What are the Use of CSS Visual rules.
v. What are the Use of CSS Display and positioning?
vi. What are the Use of CSS Box and Grid model?

Step2: Ask any learner to write answers provided on flipchart/paper or blackboard.

Step3: Ask learners to discuss the provided answer and choose correct answers.

Step4: Provide expert view and clarifies ideas by using didactic materials.

Step5: Address any questions or concerns.

Step6: Ask trainees to read the key reading 2.3.1 in the trainee manual.

Points to Remember

- CSS (Cascading Style Sheets) is a powerful styling language used to control the presentation and layout of HTML documents. It is applied to enhance the visual appearance, layout, and responsiveness of web pages. Here are various applications of CSS.
- The version of CSS are CSS1 (Cascading Style Sheets Level 1, CSS2 (Cascading Style Sheets Level 2, CSS2.1, CSS3 (Cascading Style Sheets Level 3), CSS4).
- The types of CSS style are (Inline Styles, Internal Styles (Embedded Styles), External Styles, CSS Selectors etc.



Practical Activity 2.3.2: Applying CSS



Notes to the trainer

- This activity should take place in a computer lab where trainees should be designing a personal portfolio website to showcase your projects and skills
- Avail computer connected to the internet.
- Avail Web Browser



Key steps:

While delivering this activity, pass through the following steps:

Step1: Introduce the topic and ask learners do the task described below:

As an IoT web developer, you are asked to go to the computer lab for applying CSS

Step2: Explain the task and provide clear work instructions.

Step3: Demonstrate how to apply CSS. While demonstrating, explain the steps to follow

Step4: Ask learners to apply CSS and monitor the procedures.

Step5: Verify Whether the use CSS visual rules (font, colors and background colors, opacity, background image),use CSS display and positioning(relative, absolute, fixed,

sticky, block),use CSS box and grid model(heigh and width, border, border radius, padding, margin, visibility, auto) is performed

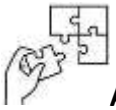
Step6: Ask trainees to read key reading 2.3.2.



Points to Remember

- Step-by-step guide for applying CSS to your web page, focusing on visual rules, display and positioning, and the box and grid model.

  Step 1: Set Up Your HTML Document
  Step 2: Use CSS Visual Rules
  Step 3: Use CSS Display and Positioning
  Step 4: Use CSS Box and Grid Model
  Step 5: Use CSS Grid Model (Optional)



Application of learning 2.3.

You are tasked with developing a **smart home IoT dashboard** that allows users to control and monitor their home devices such as lights, thermostats, and security cameras. The goal is to make the interface user-friendly, visually appealing, and responsive across various devices (smartphones, tablets, desktops).

Checklist

Criteria	Indicator	Observation	
		Yes	No
Application of CSS is performed while design an event invitation page for an upcoming family gathering or party	Versions of CSS is used		
	Style types is created		
	CSS Concepts is Applied		
	CSS Visual rules is Used		
	CSS Display and positioning is Used		
	CSS Box and Grid model is Used		



Learning outcome 2 end assessment

Theoretical assessment

Section 1: Read the following statement related to IoT Web application using PHP, and answer by True if the statement is correct or False if the statement is wrong

a. User personas are fictional representations of the ideal users for a system.

Answer:True

b. A consistent design is not necessary for a good user interface.

Answer:False

c. The tag in HTML is used to create inline elements.

Answer:False

d. CSS stands for Cascading Style Sheets and is used to style HTML elements.

Answer:True

e. The CSS position: fixed; property allows an element to stay in the same position relative to the viewport, even when the page is scrolled.

Answer:True

Section 2: Multiple Choice Questions (MCQs)

1. Which of the following is not a feature of a good user interface?

    A. Clear and intuitive navigation
    B. Consistent design
    C. Overloaded information on a single page
    D. Use of familiar UI patterns

Answer: C) Overloaded information on a single page

2. Which HTML tag is used to create an ordered list?

    A.
    B.

C.
D. <dl>

Answer: B)

3. Which of the following CSS properties is used to control the transparency of an element?

   A. opacity
   B. display
   C. position
   D. border-radius

Answer: A) opacity

4. What is the main purpose of creating wireframes during UI development?

   A. To add color and styles to the interface
   B. To outline the basic structure and layout of the user interface
   C. To write the backend logic for the application
   D. To finalize the content for the UI

Answer: B) To outline the basic structure and layout of the user interface

5. Which CSS property is used to change the font size of an element?

   A. font-size
   B. font-weight
   C. text-align
   D. font-style

Answer: A) font-size

Section 3: Matching Questions

Match the following HTML tags with their appropriate use:

1. <table>	A) Used to display tabular data.
2. <h1>	C) Used to create a form for user input.
3. <form>	B) Used to define the most important heading.

4. <a>	E) Used to display images.
5. <img>	D) Used to create hyperlinks.
6.<tr>	

Answers:

1. - A
2. - B
3. - C
4. - D
5. - E

Section 4: Open Questions

1. What is the purpose of using wireframes in UI development?

Answer:
Wireframes are used to outline the basic structure and layout of a user interface. They provide a visual guide for content placement and functionality, helping designers and developers understand the flow and arrangement of different elements on the page.

2. Explain the difference between padding and margin in CSS.

Answer:
Padding is the space between an element's content and its border, while margin is the space between the element's border and the surrounding elements. Padding affects the element's internal spacing, whereas margin affects the spacing between different elements.

Practical assessment

You are developing a smart agriculture monitoring dashboard that allows farmers to track various environmental parameters (e.g., soil moisture, temperature, humidity) across their farm using IoT sensors. The goal is to create a user-friendly, visually appealing, and responsive interface that can be accessed on multiple devices like desktops, tablets, and smartphones.

Criteria	Indicator	Observation

		Yes	No
User interface is correctly developed	UI requirements are identified (wire frame)		
	HTML tags are used		
	CSS is applied(Inline Styles, Internal Styles,external style)		

END



Further information to the trainer

mohammad el-basioni, basma m., and sherine m. abd el-kader. "designing and modeling an iot-based software system for land suitability assessment use case." environmental monitoring and assessment 196.4 (2024): 380.

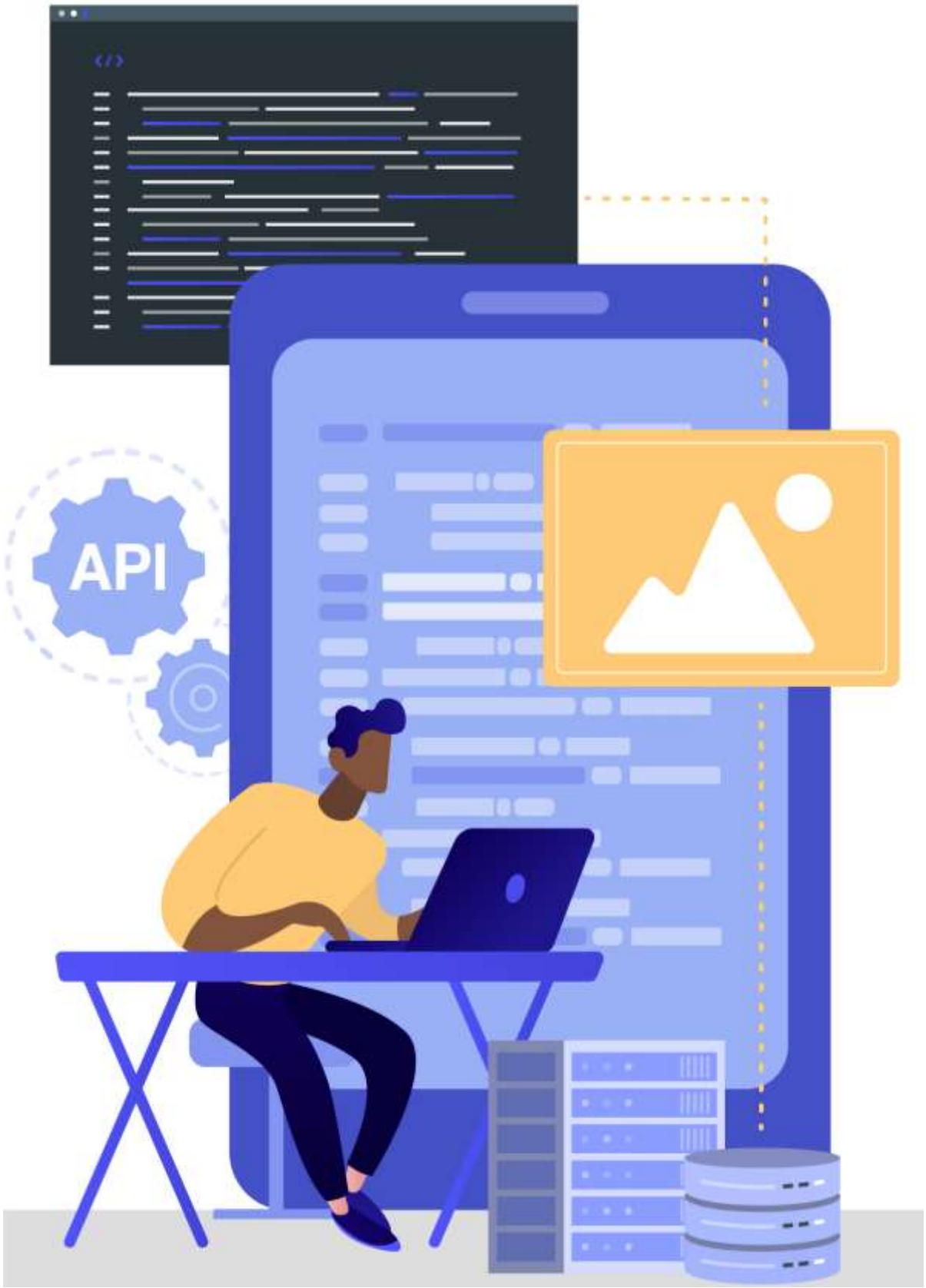
https://www.w3schools.com/html/html_intro.asp

sinlae, fried, et al. "penggunaan framework laravel dalam membangun aplikasi website berbasis php." jurnal siber multi disiplin 2.2 (2024): 119-132.

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API

Indicative contents
3.1 Interpret API Endpoints **3.2 Use API data on User Interface** **3.3 Document IoT Web Application**

Key Competencies for Learning Outcome 3: Integrate API Endpoints with User Interface.

● Knowledge	● Skills	● Attitudes
● Description of endpoint HTTP Methods ● Description the typical structure of an API response	● Interpreting API Endpoints ● Using API data on User Interface ● Documenting IoT Web Application	● Being meticulous in checking the structure of API requests ● Having a problem-solving mindset to enables you effectively analyse and resolve API issues. ● Adaptability and Continuous Learning for integrating APIs and enhancing user interfaces as technologies ● Having collaboration and feedback when documenting API Endpoints with User Interface.



Duration: 20hrs

Learning outcome 3 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Describe clearly endpoints HPPT methods based on the user interface in line with http communication protocol
2. Interpret correctly interface (API) endpoints based on to API documentation.
3. Use properly Interface (API) data the user interface in line with http communication protocol
4. Document effectively the report based on system functionality



Resources

Equipment	Tools	Materials
• Computer • Projector	• Text editor • Webserver • Web browser • DBMS	• Electricity • Internet



Advance Preparation:

Before delivering this learning outcome, you are recommended to:

- Avail prepared computer lab

Indicative content 3.1: Interpret API Endpoints

Duration: 4hrs

Theoretical Activity 3.1.1: Description of Endpoint HTTP Method

Notes to the trainer:

- ✓ Trainer may use small group to describe endpoint HTTP method
- ✓ Avail of computer connected to the internet

Key steps:

While delivering this activity, pass through the following steps:

Step 1: Introduce the activity and request learners to respond to the following questions:

i. What are HTTP method API endpoint?
ii. How can you check URL structure in API endpoint?
iii. What are the types of request parameter used in API endpoint?
iv. Provide the typical structure of an API response
v. What are different types of API authentication
vi. What are error codes and error response format?

Step 2: Ask any learner to write answers provided on flipchart/paper or blackboard.

Step 3: Ask learners to discuss the provided answer and choose correct answers.

Step 4: Provide expert view and clarifies ideas by using didactic materials.

Step 5: Address any questions or concerns.

Step 6: Ask trainees to read the key reading 3.3.1 in the trainee manual.

Points to Remember

- The different HTTP methods used in API endpoints are :(GET: Retrieves data from the server. POST: Sends new data to the server to create a new resource.PUT: Updates an existing resource with new data.PATCH: Partially updates an existing resource.DELETE: Removes a resource from the server.)

- the types of request parameters used in API endpoints are: (Path Parameters, Query Parameters, Body Parameters, Header Parameters)
- The different types of API authentication are (API Key Authentication, OAuth, Basic Authentication, Bearer Token, Digest Authentication)
- to check error codes and error response format use (Check the Status Code, Inspect the Response Body, Test Edge Cases).



Practical Activity 3.1.2: Interpreting API HTTP Method



Notes to the trainer

- ✓ Trainer may use small group to describe endpoint HTTP method
- ✓ Avail computer connected to the internet



Key steps:

While delivering this activity, pass through the following steps:

Step1: Introduce the activity by asking trainees to answer the following questions:

As an IOT developer, you are asked to go to the computer lab to Interpret API Endpoints

Step2: Explain the task and provide clear work instructions

Step3: Demonstrate on how to Interpret API Endpoints. While explaining give the steps to follow

Step4: Asks learners to Interpret API Endpoints and monitor the procedures.

Step5: Verify whether check the structure of API responses, Check if the API requires authentication, Check error codes and error response format.

Step6: Ask trainees to read key reading 3.1.2.



Points to Remember

- **Here's are step-by-step guide to help you interpret an API:**

  Step 1: Understand API Documentation
  Step 2: Identify Endpoints
  Step 3: Determine HTTP Methods
  Step 4: Analyze Request Parameters

Step 5: Test the Endpoints
Step 6: Handle Authentication
Step 7: Implement Error Handling
Step 8: Review Rate Limits
Step 9: Build Your Application



Application of learning 3.1.

Your parent wants to find nearby restaurants using a location-based service, such as the Google Places API, while planning a weekend outing. you are required to do that task.

Checklist

Criteria	**Indicator**	**Observation**	
		Yes	**No**
Integrate API Endpoints with User Interface	API URL Structure is indicated.		
	Request Parameters is identified		
	Structure of API Responses is identified		
	Error Codes and Error Response Format are checked		
	The API Requires Authentication is checked		

Indicative content 3.2: Use API Data on User Interface

Duration: 4hrs

Theoretical Activity 3.2.1: Use API data on user interface

Notes to the trainer:

- ✓ While delivering this content, a small group can be used for introducing Use API data on user interface
- ✓ ensure you have real-world examples of APIs that students can interact with, like weather, stock, or public data APIs

Key steps:

While delivering this activity, pass through the following steps:

1. Involve trainees in group formulation

Introduce the activity and ask learners to respond to the following questions:

i. What are the URL requests?
ii. What are the HTTP request?
iii. What are Testing and Debugging techniques used to identify the Use of API data on User Interface
iv. What is data rendering?
v. Explain the task and provide clear work instructions.
vi. Make an explanation on URL requests and HTTP request. While explaining give the steps to follow

2. Asks learners for to work on URL requests and HTTP request and monitor the procedures

3. Ask trainee to write their answers on paper, blackboard, flipchart or white board

5.Ask question for clarification if any.

6. Ask trainees to read key reading 3.2.1

Points to Remember

- The URL requests (Endpoints Method Headers Data (or body), Status (code and message))
- HTTP request (POST requests to create records, GET request to read or get a resource from the server, PUT and PATCH requests to update records, DELETE request to delete a resource from a server)
- Testing and Debugging techniques used to identify the Use of API data on User Interface used unit testing,intergration testing,end to end testing,manual testing
- Data rendering in the context of using API data on a user interface (UI) refers to the process of taking data retrieved from an API and displaying it in a meaningful and user-friendly way on a web or mobile application.



Practical Activity 3.2.2: Use API Data on User Interface



Notes to the trainer

- This activity should take place in a computer lab where trainees Use API data on User Interface
- Avail text editor
- Avail browser
- computers



Key steps:

While delivering this activity, pass through the following steps:

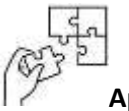
1. Introduce the topic and ask trainees do the task described below:
   You are requested to go to the computer lab where trainees Use API data on User Interface
2. Explain the task and provide clear work instruction.
3. Demonstrate how to use API data on user Interface. While demonstrating, explain the steps to follow.
4. Ask learners to use API data on user Interface and monitor the procedures.

5. Verify whether URL requests (Endpoints, Method, Headers, Data (or body), Status (code and message)) is correctly performed
6. Ask trainees to read key reading 3.2.2
7. Ask trainees to perform the task provided in application of learning 3.2.2



Points to Remember

Step 1: Understand the API
Step 2: Set Up Your Development Environment
Step 3: Create the User Interface
Step 4: Implement API Calls
Step 5: Handle Responses
Step 6: Render Data in the UI
Step 7: Implement User Interactions
Step 8: Optimize Performance
Step 9: Test the Integration
Step 10: Deploy the Application



Application of learning 3.2.

You are tasked with building a weather dashboard that displays the current weather and forecast for a selected city. The app retrieves data from a public weather API, allows users to search for cities, and displays the weather data dynamically. You will handle various API requests, ensure correct data rendering, and make sure the dashboard is responsive

Checklist

Criteria	Indicator	Observation	
		Yes	No
Use API data on User Interface	URL Requests is indicated		
	HTTP Requests is applied		
	Testing and Debugging techniques is applied		
	Data Rendering on the User Interface is used		

Indicative content 3.3: Document IoT Web Application

Duration: 2hrs

Theoretical Activity 3.3.1: Document IoT web application

Notes to the trainer:

- While delivering this content, a small group can be used for Documenting IoT Web Application.
- Prepare Clear Template for User Guide Documentation
- Guidelines for Report Writing
- Sample Bills of Quantities

Key steps:

While delivering this activity, pass through the following steps:

1. Introduce the activity and request learners to respond to the following questions
   i. What is User Guide Documentation in IoT web application?
   ii. How to do Report of Work Done in IoT web application?
   iii. Provide Bills of quantities
   iv. Explain the task and provide clear work instructions
   v. Make an explanation on documentation of IoT web application. While explaining give the steps to follow.
2. Ask learners for to make documentation of IoT Web application and monitor the procedures.
3. Verify whether User guide documentation, Report on work done, Provide Bills of quantities is performed
4. Ask trainee to write their answers on paper, blackboard, flipchart or white board
5. ask question for clarification if any.
6. Ask trainees to read key reading 3.3.1.

Points to Remember

- In IoT web application User Guide Documentation is to Create detailed user manuals or guides that explain how to set up, use, and troubleshoot the IoT web application. This should include step-by-step instructions, screenshots, and any technical requirements
- To do Report on Work Done You Should Prepare a comprehensive report documenting the development process of the IoT web application. It should include the project scope, milestones achieved, challenges encountered, and solutions implemented
- The things that Provide Bills of Quantities (BoQ) required hardware, software, and other materials used in the development and deployment of the IoT system, along with their costs and quantities.



Practical Activity 3.3.2: Document IoT web application

Notes to the trainer

- This activity should take place in a computer lab where trainees make documentation of IoT Web Application
- Avail text editor
- Avail browser
- computers

Key steps:

While delivering this activity, pass through the following steps:

1. Introduce the topic and ask trainees do the task described below:

   i. You are requested to go to the computer lab where trainees to make Documentation of IoT web application

2. Explain the task and provide clear work instruction.

3. Demonstrate how to document IoT Web Application. While demonstrating, explain the steps to follow.

4. Ask learners to make Document IoT Web Application and monitor the procedures.

5. verify whether User guide documentation, Report on work done, Provide Bills of quantities is performed

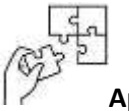
6. Ask trainees to read key reading 3.3.2

7. Ask trainees to perform the task provided in application of learning 3.3.2



Points to Remember

Step 1: Introduction
Step 2: Prerequisites
Step 3: System Architecture
Step 4: API Endpoints Documentation
Step 5: User Interface Integration
Step 6: Setup and Installation
Step 7: Usage Instructions
Step 8: Troubleshooting
Step 9: Maintenance and Updates
Step 10: Conclusion
Step 11: Appendices
Step 12: Review and Feedback



Application of learning 3.3.

A student buys a new smart home device (like a smart light bulb or thermostat) and is responsible for creating a user guide for their family. You are required to Set Up their Smart Home Device so, you are able to Write a simple user guide to help others set up and control the device through the app, Documenting the steps taken during setup, noting any issues and solutions as a report on work done and Estimating the cost of all devices and accessories involved in the setup, like Wi-Fi extenders or smart plugs, as a basic Bills of Quantities

Checklist

Criteria	Indicator	Observation	
		Yes	No
Document IoT Web Application	User guide documentation is established		
	Report on work done is used?		
	Bills of Quantities is mentioned?		



Learning outcome 3 end assessment

Theoretical assessment

A. Multiple Choice Questions (MCQs):

1. Which HTTP method is used to retrieve data from a server?
    a. POST
    b. PUT
    c. GET
    d. DELETE

Answers: c

2. What does an API response typically contain?
    a. Status Code, Headers
    b. Status, Data, Error
    c. Request Body, Query Parameters
    d. URL, Endpoint

Answer: b

3. Which HTTP method is used to update an existing resource?
    a. A) GET
    b. B) POST
    c. C) DELETE
    d. D) PUT

    Answer: d

4. What does the status code "404" represent in an API response?
    a. A) Resource not found
    b. B) Server error
    c. C) Unauthorized access
    d. D) OK

Answer: A

5. Which part of an API request contains data to be sent to the server?
    a. A) Headers
    b. B) Body
    c. C) URL
    d. D) Endpoint

Answer: B

6. What is the purpose of an API key?
    a. A) It defines the structure of the response
    b. B) It authenticates requests made to the API
    c. C) It sets the timeout for a request
    d. D) It specifies the data type of the response

Answers: B

7. Which method is used to delete a resource from the server?
    a. A) PATCH
    b. B) GET
    c. C) POST
    d. D) DELETE

Answer: D

B. Read the following statement related to IoT Web application using PHP, and answer by True if the statement is correct or False if the statement is wrong

1. A DELETE request is used to create new resources.

Answer:False

2. API keys are a form of API authentication.

Answer:True

3. A PATCH request is used to partially update an existing resource.

Answer:True

4. An API with status code 200 means the request has failed.

Answer:False

5. API endpoints usually require authentication to secure access to data.

Answer:True

6. In a REST API, POST is the preferred method for retrieving data.

Answer:False

7. The HTTP DELETE method is used to modify an existing resource.

Answer:False

C. Read the following statement related to IoT Web application using PHP and write the letter of description that corresponding to the correct http method:

Answer	http method	Description
.....	1. GET	a) Retrieves data from the server
.....	2. POST	b) Creates new resources
.....	3. PUT	c) Updates an existing resource
.....	4. DELETE	d) Deletes a resource

ANSWERS

1.A

2.B

3.C

4.D

D. Open-Ended Questions

1. Explain the steps to integrate API data into a user interface, and how you would handle errors during the process

Answer: To integrate API data into a user interface, follow these steps:

1. Set up the endpoint: Identify the API endpoint and the necessary HTTP method.
2. Send the request: Use fetch or axios to send the request to the API.
3. Handle the response: Parse the JSON response and extract the necessary data.
4. Render the data: Use the extracted data to update the UI.
5. Error Handling: Implement error handling by using try...catch blocks. If an error occurs, display a user-friendly message depending on the error code (e.g., show "City not found" for a 404 error).

2. Explain the role of authentication in securing API endpoints. How would you implement API key authentication in a project?

Answer: Authentication secures API endpoints by verifying the identity of users or applications attempting to access them. To implement API key authentication:

1. Obtain an API key: Sign up for the API service and obtain a unique API key.
2. Include the API key in requests: Add the API key to the request headers or as a query parameter.
3. Validate the API key on the server: The server checks the provided key against a database of valid keys before granting access to the requested resources.

3. Describe the steps for handling API errors in a user interface. How would you display different error messages to users based on the error codes (e.g., 400, 401, 404, 500)?

   Answer:

   - ✓ Check response status: After making an API request, check the response status code.
   - ✓ Define error messages: Create a mapping of status codes to user-friendly error messages.
   - ✓ Display messages: Based on the status code, display the corresponding message on the user interface. For instance:
   - ✓ 400: "Bad request. Please check your input."
   - ✓ 401: "Unauthorized. Please log in."
   - ✓ 404: "Not found. Please verify the requested resource."
   - ✓ 500: "Server error. Please try again later."

4. Walk through the process of making a GET request in a user interface. Include all necessary steps such as setting up the endpoint, sending the request, handling the response, and rendering the data on the page.

   Answer:

1. Set up the endpoint: Define the API URL, e.g., https://api.example.com/data.
2. Send the GET request: Use fetch:

a. fetch('https://api.example.com/data')
i. . then (response => response.json())
ii. . then (data => renderData(data))
iii. . catch (error => handleError(error));

3. Handle the response: Process the received data in the renderData function.
4. Render the data: Update the HTML elements with the data values.
5. Display error messages if any occur during the request.

5. Explain how you would implement pagination and infinite scrolling when rendering data retrieved from an API. What would be the key considerations to ensure a smooth user experience?

Answer:

- Pagination:

1. Determine how many items to show per page.
2. Use query parameters (e.g., page=1) to fetch data for each page.
3. Create UI controls (e.g., "Next" and "Previous" buttons) for navigation.

- Infinite Scrolling:

1. Detect when the user scrolls to the bottom of the page.
2. Automatically fetch the next set of data when this occurs.

- Key Considerations:

1. Provide visual indicators (e.g., loading spinners) while fetching more data.
2. Handle cases where there are no more items to load.
3. Optimize performance to prevent excessive API requests.

6.Discuss the importance of testing and debugging API requests. What tools would you use, and how would you identify and resolve issues in both request and response handling?

Answer:

Testing and debugging API requests are crucial to ensure that the application functions correctly and provides a seamless user experience. Common issues include incorrect endpoints, wrong parameters, and unexpected response formats.

Tools:

Postman: For testing API endpoints and visualizing responses.

Browser Developer Tools: To inspect network requests and responses.

Logging: Implementing logging in your application can help track requests and identify errors.

Identifying and Resolving Issues:

Monitor console logs for error messages.

Compare actual responses against expected formats to catch discrepancies.

Use unit tests and integration tests to validate that API interactions work as intended.

Practical assessment

You are tasked with developing a simple weather application that fetches current weather data from a public weather API. The application will display weather information based on the user's location and allow users to search for weather data by city. Your application should handle API requests, render the data on the user interface, and provide appropriate error messages. You will also document the process and provide a report on your work.

Checklist

Criteria (	**Indicator**	**Observation**	
		Yes	**No**
API Endpoints with User Interface is properly integrated	API Endpoints are interpreted		
	API data on User Interface are used		
	IoT Web Application is documented		



Further information to the trainer

Kumar, S. P. ((April 2018)). 2. S Praveen Kumar. (April 2018). IoT-based temperature and humidity monitoring system using Raspberry Pi ResearchGate

https://www.researchgate.net/publication/325172192_IoTbased_Temperature_and_Humidity_Monitoring_System_using_Raspberry_pi

Sinlae, Fried, et al. "Penggunaan Framework Laravel dalam Membangun Aplikasi Website Berbasis PHP." Jurnal Siber Multi Disiplin 2.2 (2024): 119-132.

Sinlae, Fried, et al. "Penggunaan Framework Laravel dalam Membangun Aplikasi Website Berbasis PHP." Jurnal Siber Multi Disiplin 2.2 (2024): 119-132.

https://documentation.mindsphere.io/MindSphere/howto/howto-integrate-api.html#:~:text=A%20user%20can%20develop%20backend,functional%20logic%20in%20multiple%20applications .

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